

Toshinori Kitamura

✉ toshinori-k@weblab.t.u-tokyo.ac.jp | 📞 syuntoku14

Education

University of Alberta

2025-2028

POSTDOCTORAL RESEARCHER

- **Postdoctoral Researcher (2025):** The University of Tokyo
- **Ph.D. in Engineering (2022-2025):** The University of Tokyo
- **Master's Degree (2020-2022):** Nara Institute of Science and Technology (NAIST)
- **Bachelor's Degree (2016-2020):** Keio University, *Faculty of Science and Technology (System Design Engineering)*
- **Exchange Student (2018-2019):** University of California, Davis

Publications

Recent Working Drafts

- “Revisiting Subgradient Dominance in Robust MDPs: Counterexamples, Hardness, and Sufficient Conditions”
 - **Toshinori Kitamura**, Arnob Ghosh, Alex Ayoub, Thang D. Chu, Csaba Szepesvári, Arxiv 2026

Conference Proceedings

- “Provably Efficient RL under Episode-Wise Safety in Constrained MDPs with Linear Function Approximation”
 - **Toshinori Kitamura**, Arnob Ghosh, Tadashi Kozuno, Wataru Kumagai, Kazumi Kasaura, Kenta Hoshino, Yohei Hosoe, Yutaka Matsuo, **Neurips 2026 Spotlight**
- “Near-Optimal Policy Identification in Robust Constrained Markov Decision Processes via Epigraph Form”
 - **Toshinori Kitamura**, Tadashi Kozuno, Wataru Kumagai, Kenta Hoshino, Yohei Hosoe, Kazumi Kasaura, Masashi Hamaya, Paavo Parmas, Yutaka Matsuo, **ICLR 2025**
- “Regularization and Variance-Weighted Regression Achieves Minimax Optimality in Linear MDPs: Theory and Practice”
 - **Toshinori Kitamura**, Tadashi Kozuno, Yunhao Tang, Nino Vieillard, Michal Valko, Wenhao Yang, Jincheng Mei, Pierre Ménard, Mohammad Gheshlaghi Azar, Rémi Munos, Olivier Pietquin, Matthieu Geist, Csaba Szepesvári, Wataru Kumagai, Yutaka Matsuo, **ICML 2023**
- “Geometric Value Iteration: Dynamic Error-Aware KL Regularization for Reinforcement Learning”
 - **Toshinori Kitamura**, Lingwei Zhu, Takamitsu Matsubara, **ACML 2021**
- “Cautious Actor-Critic: Stable off-policy deep reinforcement learning for continuous control”
 - Lingwei Zhu, **Toshinori Kitamura**, Takamitsu Matsubara, **ACML 2021**

Preprint, Domestic Conference, and Workshop Papers

- “A Policy Gradient Primal-Dual Algorithm for Constrained MDPs with Uniform PAC Guarantees”
 - **Toshinori Kitamura**, Tadashi Kozuno, Masahiro Kato, Yuki Ichihara, Soichiro Nishimori, Akiyoshi Sannai, Sho Sonoda, Wataru Kumagai, Yutaka Matsuo, RLC Safety WS 2024 (oral)
- “On Various Problem Settings of Sequential Decision Making and How Prior Knowledge of Problems Affects Theoretical Guarantees”
 - Tadashi Kozuno, **Toshinori Kitamura**, Yuki Ichihara, Makoto Hagiwara, JSAI 2023
- “KL-Entropy-Regularized RL with a Generative Model is Minimax Optimal”
 - Tadashi Kozuno, Wenhao Yang, Nino Vieillard, **Toshinori Kitamura**, Yunhao Tang, Jincheng Mei, Pierre Ménard, Mohammad Gheshlaghi Azar, Michal Valko, Rémi Munos, Olivier Pietquin, Matthieu Geist, Csaba Szepesvári, Arxiv 2022
- “ShinRL: A Library for Evaluating RL Algorithms from Theoretical and Practical Perspectives”
 - **Toshinori Kitamura**, Ryo Yonetani, NeurIPS RL Workshop 2022

Experience

Research

- **RIKEN Center for Advanced Intelligence Project Deep Learning Theory Team (2023-)**
 - Statistical RL theory research: constrained RL and general function approximation.
- **The University of Tokyo Matsuo Lab (2022-)**
 - Statistical RL theory research: constrained RL, general function approximation, KL-Entropy regularization, etc.
- **NAIST Robot Learning Lab (2020-2022)**
 - Research on error propagation analysis of KL-Entropy regularized DP.
- **OMRON SINIC X Corporation (2021-2022, 2023-2024)**
 - Research on RL library with JAX & statistical RL theory
- **National Institute of Advanced Industrial Science and Technology (2019-2020)**
 - RL application research for mobile robot in human crowds.
- **UC Davis Chen-Nee Chuah Lab (2019)**
 - RL application research for traffic control.

Teaching

- **Deep RL Spring Seminar:** Slides1 (DP), Slides2 (Multi-step RL)
- **Deep Learning Course**

Working

- Robotics: Integral AI, Inc. (2022), Ugo, Inc. (2018)

Projects

<https://syuntoku14.github.io/#projects>